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UoG / UESTC Joint School of Engineering

Professional Practice : Lecture 1B

Literature search and referencing

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Adapted from Dr Derek Gray's notes



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Outline

- Literature review project
- Sources of information
- Searching
- Finding
- How not to use references
- How to use references

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Objectives

- Research a small area of specific interest
- Substantiate your written work by proper reference to published material
- Cite accurately and adequately
- Use library catalogues and other search mechanisms (internet) to find relevant material
- Write precisely, concisely, with good style and in an accepted format

Purpose of a literature search

- To identify the existing information sources most relevant to the research question
- To learn methods & approaches that are appropriate for your study
- Broaden your horizons a little…

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A thorough literature search helps to answer questions like these:

1. What research/development has already been conducted that is relevant to my research question?
2. What theories have guided the research/development that has already been conducted?
3. What hypotheses have been tested?
4. How have other researchers defined their variables?
5. What kinds of research designs have other researchers used?
6. What resource limitations did they have?
7. What mistakes have other researchers made?
8. What suggestions have been made for future research?

Information sources

- Sources
  - Books
  - Reports
  - Journals Articles
  - Popular Science Magazines
  - Websites
- Characteristics
  - Reliability
  - Availability
  - Current

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SEARCHING SKILLS

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Searching skills

- Searching techniques
- Aids to locating information
- Working with references

Searching skills

- Plan your search
- List keywords
  - synonyms
  - single/plural forms
  - abbreviations, acronyms
  - American/English spellings
- Select appropriate databases
  - User guides, on screen help

Example of a search strategy

Search: “The weathering of concrete structures”

Subject keywords:

concrete

weather, weathering, erosion

air pollution, environment, acid rain

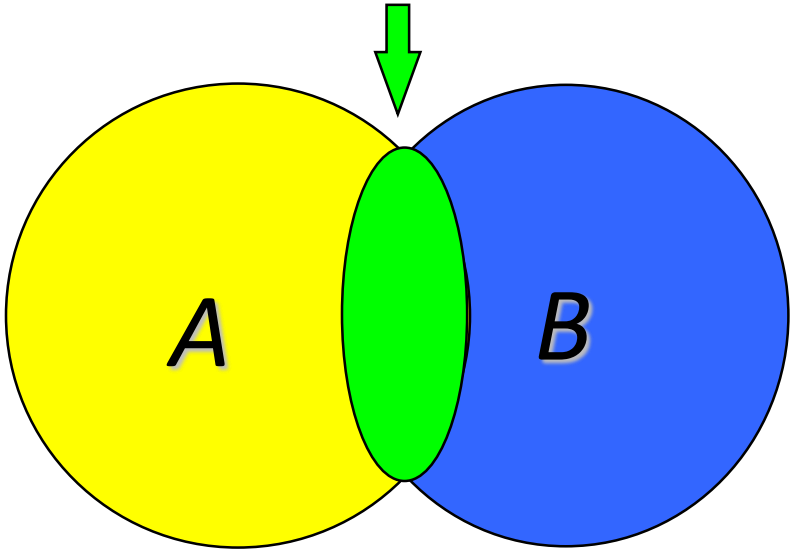
Create search statements using **Boolean** operators

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Boolean ‘AND’

‘AND’ narrows your search

References must contain both
term A AND term B

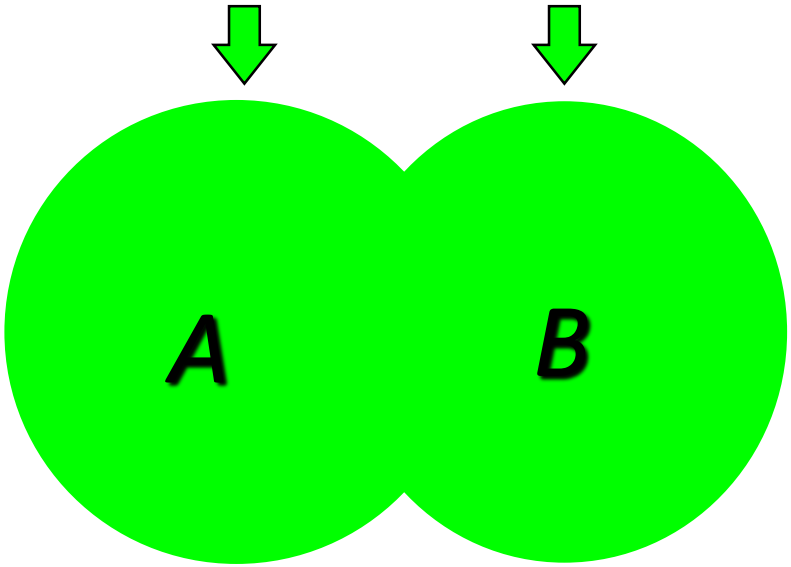


eg ‘concrete’ AND ‘erosion’

Boolean ‘OR’

‘OR’ widens your search

References may contain either A OR B

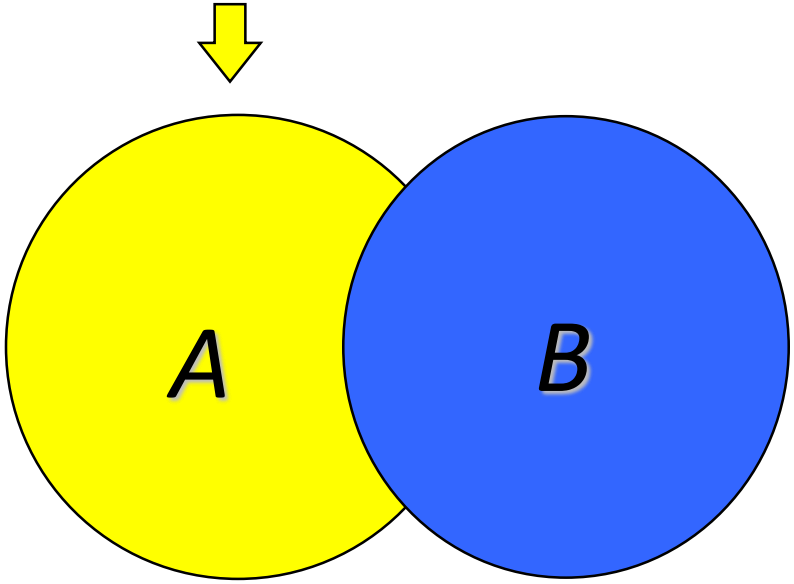


e.g. pollution OR environment

Boolean ‘NOT’

‘NOT’ narrows your search

References must contain term A but NOT term B



eg air NOT water

Examples

- concrete
- weather* or erosion
- 1 AND 2

AND

- pollution or environment

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Examples

- surface mining or open cast mining
- concrete bridg*
- gps or global positioning system
- sulphur or sulfur
- concrete bridg* AND (failure* OR collapse)
- concrete AND (weather* OR erosion) AND (pollution or environment*)

Locating materials

- Books, conferences, bibliographies, theses, reports
  - Library at UESTC http://www.lib.uestc.edu.cn/
  - Library at UoG https://www.gla.ac.uk/myglasgow/library/
  - IEEE database https://ieeexplore.ieee.org/Xplore/home.jsp.
  - Wiley database https://onlinelibrary.wiley.com/
  - IntechOpen https://www.intechopen.com/
  - And more:
    - http://www.wpmc-archives.com/site/
    - http://ap-s.ei.tuat.ac.jp/isapx/index.html
    - http://ci.nii.ac.jp/
    - http://worldwide.espacenet.com/advancedSearch?locale=en_EP
    - http://ntrs.nasa.gov/search.jsp
    - http://www.dtic.mil/dtic/
- You can consult the list of books and journals the libraries have access to and the web links to resources

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Report series

- Significant source of engineering information
- Series include
  - AGARD (Advisory Group for Aerospace Research & Development)
  - BRE (Building Research Establishment)
  - CIRIA (Construction Industry Research & Information Association)
  - NASA (National Aeronautics & Space Administration)
  - TRL (UK's Transport Research Laboratory)
  - TRB (USA's Transportation Research Board)
- Again see what the libraries have access to

General databases for Engineers

- ANTE (Abstracts in New Technologies & Engineering)
- Ei Compendex (EDINA)
- INSPEC (EDINA)
- Web of Science (ISI Science Citation Index)
- see Library website for more specific titles
- various electronic journal packages

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Searching summary

- Define a search strategy
- Use the Library websites for pointers to key resources both with the libraries and on the web.
- Use the library websites to know if your library holds a copy either e-copy or physical copy.

How to use references - Plagiarism

- 'Plagiarism' can be defined as the attempt to gain advantage for yourself – academic advantage, financial advantage, professional advantage, advantage of publicity – by trying to fool someone, such as a teacher, an editor, an employer, or a reader, into thinking that you wrote something, thought something, or discovered something which, in actual fact, someone else wrote, thought, or discovered. Plagiarism is sometimes defined, aphoristically, as 'literary theft'. (McNaughton, 1995).

Wind Energy - Microsoft Internet Explorer

View Favorites Tools Help

Search Favorites

http://www.eere.energy.gov/RE/wind.html

U.S. Department of Energy
Energy Efficiency and Renewable Energy
Bringing you a prosperous future where energy is clean, abundant, reliable, and affordable

EERE Home

Home | Site Map | A-Z Index | EERE Information Center

Search

Wind



Wind & Hydropower Technologies Program Web Site

Wind Organizations & Resources

- U.S. Department of Energy
- U.S. Federal Government
- U.S. State & Local
- U.S. Trade & Nonprofit
- U.S. Universities and Research Institutes
- International Organizations
- Discussion Groups
- Newsletters & Magazines
- Products & Services

Wind Energy Topics

Wind energy uses the energy in the wind for practical purposes like generating electricity, charging batteries, pumping water, or grinding grain. Large, modern wind turbines operate together in wind farms to produce electricity for utilities. Small turbines are used by homeowners and remote villages to help meet energy needs.

More basic information about wind energy is also available.

Wind Energy Technologies

Modern wind turbines are divided into two major categories: horizontal axis turbines and vertical axis turbines. Old-fashioned windmills are still seen in many rural areas.

Wind Turbine Use

Wind turbines are used around the world for many applications. Wind turbine use ranges from homeowners with single turbines to large wind farms with hundreds of

Wind News

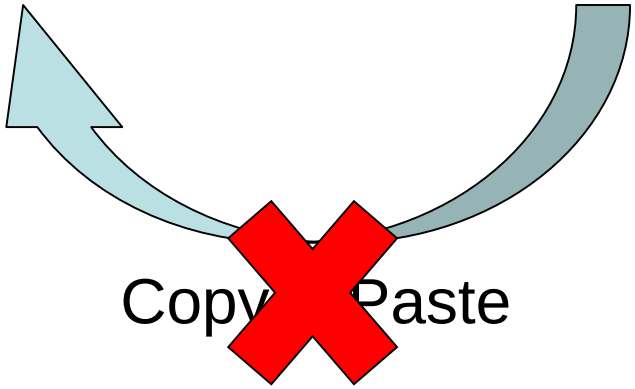
- Colorado Utility Seeks 500 Megawatts of Renewable Energy
  September 01, 2004
- EIA: U.S. Renewable Energy Use Up 3 Percent in 2003
  August 25, 2004
- California Approves New Transmission Line for Wind
  August 18, 2004

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How to use references - Plagiarism

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How to use references - **Plagiarism**

Wind energy uses the energy in the wind for practical purposes like generating electricity, charging batteries, pumping water, or grinding grain. Large, modern wind turbines operate together in wind farms to produce electricity for utilities. Small turbines are used by homeowners and remote villages to help meet energy needs.

References:

http://www.eere.energy.gov/RE/wind.html

How to use references – **Poor use**

"Wind energy uses the energy in the wind for practical purposes like generating electricity, charging batteries, pumping water, or grinding grain. Large, modern wind turbines operate together in wind farms to produce electricity for utilities. Small turbines are used by homeowners and remote villages to help meet energy needs." [1]

References

[1] http://www.eere.energy.gov/RE/wind.html

How to use references – Acceptable use

The kinetic energy in the wind can be harnessed for mechanical tasks such as pumping water, or grinding grain, or for producing electricity. For electricity generation, groups of wind turbines are often used, while a single small turbine may be used to meet the energy requirements of a single dwelling or a remote village. [1]

References

[1] http://www.eere.energy.gov/RE/wind.html

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How to use references – **Good use**

The kinetic energy of the wind can be harnessed for a range of applications. These include mechanical power generation for pumping water or grinding grain as well as generating electricity [1]. Large wind turbines can be located together in wind farms to produce electricity either on land or offshore [2]. Single, smaller wind turbines can be used to meet the energy requirements of remote villages or even individual dwellings [3].

References

[1] http://www.eere.energy.gov/RE/wind.html

[2]……

Referencing

- Endnote – https://endnote.com/
- Mendeley - https://www.mendeley.com/
- Bibtex - http://www.bibtex.org/

- Use the IEEE referencing format – Mendeley and Endnote should have those styles embedded in their Word addons
  - Guide
  - Adding a reference style to Word processor



IEEE2006OfficeOnline.xsl

Numeric IEEE system

In the text: “In a recent study, Smith [2] argued that…”

In the bibliography:

[2] D. Smith, “Heavy metals in landfill leachate,” *Resource Conservation*, vol. 23, num. 10, pp.3-8, June 1998.

Use a hanging indent, set this with the Paragraph *formatting*

Author details

- Put the initials first, followed by the surname as they appear in the publication.

- Cite up to 3 author names. If there are 4 or more names cite the first author only and then “et al.”

- Put “ed(s)” after the surname if appropriate.

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Title details

- Give the full title including any subtitle separated by a colon
- Only capitalise the first word
- Book/periodical titles should be in *italics*
- Periodical titles can be shortened by using a recognised system of abbreviations

Other details

- Place of publication:

only cite one & include the country if not obvious

- Numeration:

volume, part/issue/number (if used by the publisher), pp. page numbers.

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Books

Author(s), Date. *Title.* Edition. Place of publication: Publisher.

[2] W.-K. Chen, *Linear Networks and Systems.* Belmont, CA: Wadsworth, 1993, pp. 123–135.

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Articles in periodicals

Author(s), Title of article. *Title of periodical,* volume, (part/issue), pages, date.

[7] J.U. Duncombe, “Infrared navigation—Part I: An assessment of feasibility,” *IEEE Trans. Electron Devices*, vol. ED-11, pp. 34–39, Jan. 1959.

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Article in a book

Author(s) of article, Date. Title of article *in:* Author(s) of book. *Title of book,* Edition. Place of publication: Publisher, Date, pages.

[3] G.O. Young, "Synthetic structure of industrial plastics," in *Plastics*, 2nd ed. vol. 3, J. Peters, Ed. New York: McGraw-Hill, 1964, pp. 15–64.

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Conference papers

Author(s) of paper, Title of paper, *Title of conference,* Place, Publisher, Date, volume, pages.

[6] S.P. Bingulac, "On the compatibility of adaptive controllers," in *Proc. 4th Annu. Allerton Conf. Circuits and Systems Theory*, New York, June 1994, pp. 8–16.

[7] J.G. Kreifeldt, "An analysis of surface-detected EMG as an amplitude-modulated noise," presented at the *1989 Int. Conf. Medicine and Biological Engineering*, Chicago, IL., June 1989.

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Theses

Author, *Title.* Degree. Awarding body , date.

[3] J. Williams, “Narrow-band analyzer,” Ph.D. dissertation, Dept. Elect. Eng., Harvard Univ., Cambridge, MA, 1993.

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Electronic journal article

Author(s). (Year). Title of article. *Journal title* [Type of medium], volume (issue). Available at: URL [Access date]

[2] R.J. Vidmar. (1992, August). On the use of atmospheric plasmas as electromagnetic reflectors. *IEEE Trans. Plasma Sci.* [Online]. *21(3).* pp. 876–880. Available at: http://www.halcyon.com/pub/journals/21ps03-vidmar [accessed 21st October, 2013]

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Summary

- Have you recorded enough information to fully identify the item?
- Have you adopted the standardised IEEE form of citation?
- Have you been consistent?
- An automated referencing tool will help in keeping it tidy and consistent.